



Drug Safety & Regulatory Intelligence Platform

Pharmacovigilance Signal Detection

ICH M4 CTD Gap Analysis

Built with IBM Bob

THE CHALLENGE

The Problem We Solve



Signal Detection is Manual & Slow

Pharmacovigilance teams manually sift through thousands of adverse event reports needing specialist statistical expertise (PRR, ROR, Chi-square).

6–18 months

Average signal detection backlog



CTD Gaps Found After Submission

Regulatory dossier managers miss ICH M4 CTD structure gaps by hand — leading to Major Objections from health authorities and costly delays.

\$300M–\$800M

Cost of failed regulatory submission



Expensive Licensed Software

Small biotech and generic manufacturers cannot afford Oracle Argus or VigiBase. No accessible, open alternative exists.



Data Privacy Concerns

Pharmaceutical AE data is highly sensitive. Uploading to cloud-based tools creates regulatory and compliance risks for organisations.

OUR SOLUTION

What We Built

Sentinel is a fully client-side React web application that automates both pharmacovigilance signal detection and ICH M4 CTD gap analysis in one unified dashboard — running entirely in the browser with **zero backend**, zero infrastructure, and zero data privacy risk.

PRR · ROR · Chi-square · 95% CI

ICH M4 CTD Fuzzy Gap Analysis

Zero Backend · Privacy First

50,000 AE Reports in Seconds

60-Item CTD Reference Check

Built entirely with IBM Bob

CAPABILITIES

Key Features

1

Signal Detection Engine

PRR, ROR, Chi-square (Yates), 95% CI — ranks every drug–event pair as Critical / Signal / Watch / No Signal

2

CSV Upload & Validation

Accepts up to 50,000 AE report rows with column validation and clear inline error messages

3

ICH M4 CTD Gap Analysis

Levenshtein fuzzy matching against 60-item CTD reference with per-module readiness scores

4

Interactive Visualisations

Scatter plots, trend charts, bar charts, animated SVG readiness gauge powered by Recharts

5

Dark Mode + Responsive UI

Full dark/light toggle, custom Tailwind sidebar, accessible and mobile-friendly layout

6

Zero Backend / Privacy First

All computation runs in-browser — no API calls, no server, no data ever leaves the machine

TECHNOLOGY

Tech Stack

LANGUAGES

JavaScript ES2022 JSX

CSS3

FRAMEWORKS

React 18 Vite 5

Tailwind CSS 3

IBM TECHNOLOGIES

IBM Bob IBM Bob Agent Mode

IBM Bob Skill System

LIBRARIES & DEPLOY

Recharts 3 PapaParse 5

Lucide React Vercel

ARCHITECTURE

How It Works



TRY IT NOW

Live Demo

 LIVE APP

<https://bob-ai-hackathon-bob-crew.vercel.app>

 GITHUB REPOSITORY

[github.com/bhumii031/
bob-ai-hackathon-BobCrew](https://github.com/bhumii031/bob-ai-hackathon-BobCrew)

 DEMO VIDEO

See demo/demo-video-link.txt in repository

 SCREENSHOTS

See demo/screenshots/ in repository

 No login required · No setup · Opens instantly in any browser

OUR ACHIEVEMENT

What We're Most Proud Of

**End-to-end PV math in pure JavaScript**

PRR, ROR, Yates Chi-square, 95% CI — implemented from scratch, no external stats library, statistically equivalent to VigiBase

**Zero-backend, privacy-first architecture**

All signal detection and CTD gap analysis runs entirely client-side — no data ever leaves the user's browser

**Severity-weighted signal scoring**

Outcome weighting (death=3, serious=2) elevates fatal events to Critical Signal — matching real-world PV practice

**Production-quality UI in a hackathon timeframe**

Dark mode, animated SVG gauge, sortable tables, scatter plots, trend charts — all cohesive and accessible

**Built entirely with IBM Bob Agent Mode**

Every line of code generated by IBM Bob — demonstrating its capability for domain-expert life sciences applications